

Premise: I analyzed lyrics written by Olivia Rodrigo and Gracie Abrams based on how well they fit the seven basic plot archetypes, and recorded the data. My goal is to determine which archetypes are the best classifiers of each artist, and for each individual album I reviewed in the data.

Implementation Plan: I set up a logistic regression model to classify the songs as either written by Olivia Rodrigo (class 0) or Gracie Abrams (class 1). The features recorded for each song were the rating of fit of the lyrics to each of the seven plots, on a scale of 0-3. These features were scaled to follow a standard normal distribution so the magnitudes of the coefficients could be compared. Since this is a binary classifier, features with large positive coefficients are good indicators of Gracie Abrams, and features with large negative coefficients are good indicators of Olivia Rodrigo.

However, I also want to set up a logistic regression model to classify the songs according to the albums they belong to. A multinomial softmax model is ideal for this task because the dataset is small (around 90 total songs) and the classes are mutually exclusive (each song belongs to only one album). I will have to compare the values of the features for all six sets of coefficients, looking for the highest positive coefficients to mean the song has the best chance of being in that album..